

Constants

c	$3.00 \times 10^8 \text{ m/s}$	1 nm	10^{-9} m
h	$6.63 \times 10^{-34} \text{ J}\cdot\text{s}$	1 MHz	10^6 Hz
m_e	$9.11 \times 10^{-31} \text{ kg}$	1 Hz	$1/\text{s}$
N_A	6.022×10^{23}	1 amu	$1.661 \times 10^{-27} \text{ kg}$

Ch 7. Quantum Theory & Electronic Structure

A. Electromagnetic Radiation

y-ray · X-ray · UV · visible · IR · microwave · radio

	y-ray end (left)	radio end (right)
Frequency	highest	lowest
Energy	highest — strongest	lowest — weakest
Wavelength	shortest	longest

Visible, highest E → lowest: violet · blue · green · yellow · orange · red

- Highest-energy color = **violet**, or **blue** when violet isn't an option

$$c = \lambda \nu$$
$$E = h\nu = hc/\lambda$$

B. Key Scientists

Planck	Energy emitted only in fixed amounts (quanta)
Einstein	Light arrives in packets (photons) of fixed energy
Heisenberg	Position and velocity not both knowable
Bohr	e^- occupy fixed energy levels; explains line spectra
de Broglie	Matter has wave properties, $\lambda = h/(mu)$
Pauli	No two e^- share all four quantum numbers
Hund	Orbitals fill singly, spins parallel, before pairing

C. Electron Configuration

Subshell	s	p	d	f
Max e^-	2	6	10	14
Orbitals	1	3	5	7

Ground state	Follows Aufbau exactly, no gaps
Excited state	An e^- promoted higher, leaving a gap below
Paramagnetic	At least one unpaired e^-
Diamagnetic	All e^- paired
Valence e^-	The e^- in the highest n

- Aufbau: 1s 2s 2p 3s 3p **4s 3d** 4p 5s 4d 5p 6s 4f 5d
- Exceptions: **Cr** = [Ar]**4s¹3d⁵** · **Cu** = [Ar]**4s¹3d¹⁰** (half/full d wins)

Ch 8. Periodic Properties

A. Core Concepts

Penetration	An orbital reaching into the core region is less shielded → lower energy. Within a shell s < p < d < f
Pauli exclusion	No two e^- carry the same four quantum numbers → 2 per orbital, opposite spins
Shielding	Only core (inner) e^- shield effectively; valence e^- shield each other poorly
Z_{eff}	$Z_{eff} \approx Z - (\text{core } e^-)$

B. Periodic Trends

Property	→ across period	↓ down group
Z_{eff}	increases	~constant
Atomic radius	decreases	increases
Ionization energy	increases	decreases
Electron affinity	more negative	less negative

Property	Largest	Smallest
Atomic radius	lower-left	upper-right
Ionization energy	upper-right (He highest)	lower-left
Electron affinity	halogens	noble gases ≈ 0

- A **big jump** in successive ionization energies appears once the first **core** electron is removed.

Ch 8. Periodic Properties (cont.)

C. Ion Configurations

Cation	Remove from the highest n first ; transition metals lose 4s before 3d
Anion	Add into the outer p until it reaches a noble-gas configuration

D. Families

Group	Family	Reactivity
1A	alkali metals	most reactive metals; ↑ down the group
2A	alkaline earth	reactive, less than 1A
7A	halogens	reactive nonmetals; ↓ down the group
8A	noble gases	inert — full octet

- **Diagonal rule** — an element resembles its down-right diagonal neighbor: Li~Mg, **Be~Al**, B~Si.

Ch 9. Bonding I — Basic Concepts

A. Lattice Energy (U)

Energy **released** when gaseous ions come together into the solid crystal.

$$U \propto \frac{q_1 q_2}{r_1 + r_2}$$

↑ ionic charge	↑ U — dominant, compare first
↑ ionic radius	↓ U — use only to break a tie

- **Born-Haber cycle** — a Hess's-law loop running from the elements to the solid.

B. Electronegativity & Bond Polarity

EN = an atom's ability to attract the electrons shared in a bond.

$$\Delta EN = |EN_A - EN_B|$$

ΔEN	Bond type
~0	nonpolar covalent
0.4 – 1.9	polar covalent
≥ 2.0	ionic

- EN increases up and to the right; F is highest. Noble gases excluded.
- Larger ΔEN → generally a **stronger** bond, and a stronger bond is a **shorter** bond.
- Bond order: triple > double > single (shorter and stronger).

C. Formal Charge

$$FC = (\text{valence } e^-) - [\text{nonbonding } e^- + \# \text{ bonds}]$$

Comparing structures	FCs at zero, or as close to zero as possible
A -1 is unavoidable	Put it on the more electronegative atom

D. Lewis Structures — Octet Rules

Normal octet	8 valence e^- around each atom (H wants 2)
Incomplete octet	Al, B, Be are content below 8
Expanded octet	Period 3 and below (S, P, Cl, Xe, As, I) use empty d orbitals to exceed 8
Organic groups	-OH alcohol · C=O aldehyde/ketone · -COOH carboxylic acid · -NH ₂ amine